

Static Gesture Classification — Run Report

Generated 2026-05-15 01:08:45

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 425 trials
Total training paths	1 · 425 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	164 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	stats+fft → feature dim 8856
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=340 · test=85

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (5-fold)	CV std	Hold-out test acc
LDA	0.9706	0.0132	1.0000
SVM (Linear)	0.9853	0.0093	0.9882
Random Forest	0.9912	0.0072	0.9882
Logistic Regression	0.9647	0.0118	0.9882
SVM (RBF)	0.9647	0.0176	0.9647
KNN	0.9471	0.0118	0.9647

Per-class hold-out F1

Class	LDA	SVM (Linear)	Random Forest	Logistic Regression	SVM (RBF)	KNN	Support
Double_Closed_Fist	1.000	1.000	0.963	1.000	1.000	1.000	13
Double_Nothing	1.000	0.957	1.000	0.957	0.889	0.857	12
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	12
Double_Phone	1.000	1.000	0.957	1.000	0.957	1.000	12
Double_Pistol	1.000	1.000	1.000	1.000	0.909	0.923	12
Double_Spiderman	1.000	0.960	1.000	0.960	1.000	0.960	12
macro avg	1.000	0.988	0.988	0.988	0.965	0.963	85

External test (NewTestData)

Algorithm	External test acc
SVM (Linear)	0.8611
Random Forest	0.8611
KNN	0.8611
SVM (RBF)	0.8333
Logistic Regression	0.7778
LDA	0.7778

Per-class external accuracy

Class	SVM (Linear)	Random Forest	KNN	SVM (RBF)	Logistic Regression	LDA	Support
Double_Closed_Fist	1.000	0.833	1.000	1.000	1.000	1.000	6
Double_Nothing	1.000	1.000	1.000	1.000	0.400	1.000	5
Double_Okay	0.000	0.800	0.200	0.000	0.000	0.000	5

Double_Open_Palm	1.000	0.400	1.000	1.000	1.000	0.600	5
Double_Phone	1.000	1.000	0.800	0.800	1.000	0.800	5
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	5
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	5